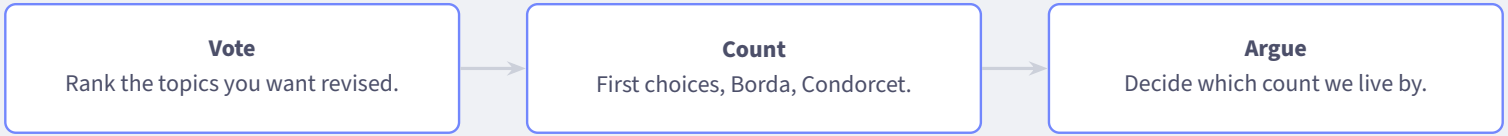
Social choice

One set of preferences, three winners



The prize is real: we revise the winner in class.

Rank all three topics in strict order, best first. No ties, and rank every topic even if you only care about one of them.

Three topics on the shortlist: $A =$ $B =$ $C =$

How they ranked them	How many said it	First choice	Borda points

Borda points for three topics: 2 for a first place, 1 for a second, 0 for a third.

Most first choices

A

B

C

Winner

Borda score

A

B

C

Winner

Head to head

A v *B*

A v *C*

B v *C*

Condorcet winner

Did all three agree?

Counting first choices looks only at _____, and throws away _____.

Borda uses _____, so a topic that is everybody's second choice can beat one that half the room ranks last.

A Condorcet winner beats every other topic _____, and sometimes _____.

stablevoting.org gave _____. Which of our three counts does that match? _____

So which method should we use for the class vote, and why?

We are revising: _____

What we just did in the room is **Question 1** on the Social Choice page, with nine students and three topics.

(a)	Rank by first choice votes and give the winner.	[2]
(b)	The Borda scores and the Borda winner.	[4]
(c)	Whether there is a Condorcet winner.	[4]
(d)	Why the methods disagree, and which you would recommend.	[5]

vknight.org/gt/topics/social-choice.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Everybody ranks the shortlist	**17.2.1**	Social Choice: Definition: Preference Function
Turning rankings into one ranking	**17.2.2**	Social Choice: Definition: A Social Welfare Function
Three counts, three winners	**17.2.3, 17.2.5, 17.2.6**	Social Choice: Simple Majority, Condorcet's Method, Borda's Method
Why no count keeps every property	**17.2.4**	Social Choice: Theorem: Arrow's Impossibility Theorem

vknight.org/gtb/chapters/social-choice

Quick quiz before you go: vknight.org/gt/topics/social-choice.html

Before the next class: read the Social Choice chapter as far as Arrow's theorem, and attempt Question 1.